

RS6240

High Performance Single-Chip 60 GHz AiP MIMO Radar Sensor

with Wireless Connectivity (BLE5.3/802.15.4/Thread/Matter)

Product Brief Rev. 1.1

Features

- **FMCW mmWave Radar**
 - 57-66 GHz Coverage With 8 GHz Continuous Sweep Bandwidth
 - 2 Transmit Channel and 4 Receive Channels with AiP Package Antenna
 - Integrated RFPLL, DPLL, Transmitter, Receiver, ADC and Baseband
 - 17.0 dBm Max EIRP Output Power per TX
 - 10 dB Typical NF per Rx
 - -95 dBc/Hz typical Phase Noise at 1MHz
 - 10/20MHz IF Bandwidth
 - Up to 800MHz/us Ramp Rate
 - HWA for 1D/2D FFT, Static Clutter Remove and CFAR Operations
- **2.4GHz Wireless Connectivity**
 - Protocols: Supports Bluetooth LE 5.3, Bluetooth Mesh, IEEE 802.15.4, Thread 1.3, Matter 1.1, and 2.4GHz proprietary
 - Bluetooth LE PHY: 1Mbps, 2Mbps, Long Range S2 (500Kbps), S8 (125Kbps)
 - Tx Max Output Power: 10dBm@BLE mode
 - Rx Sensitivity: -96dBm @BLE 1Mbps
- **Application System**
 - Two 32bit RISC-V CPU Cores with DSP, FPU (up to 256MHz) and I/D cache for Radar Post-processing, Wireless and Application
 - Security Engine: Secure Boot, Efuse Key Protection, Code encryption protection, SRAM and XIP Online Decryption, AES128/SHA/CRC/MD5/RSA/ECC/TRNG
- **Memory**
 - On Chip 551KB SRAM
 - External PSRAM support (up to 16MB)
- **Host and Device Interface**
 - x3 UART
 - x2 QSPI with XIP
 - x2 I²C
 - x8 PWM
 - x8 GPADC
 - x1 CAN-FD
 - x1 LIN
- **Clock Source**
 - 32/40MHz Crystal Oscillator for Clock Reference
 - Optional 32kHz Crystal for Low Power Operation
- **Power Management**
 - Single Power Supply from 1.7V to 5.5V
 - Built-in LDO Network for Enhanced PSRR
 - BOM-Optimized and Power-Optimized Modes
- **Low Power Property**
 - Shutdown, Standby, Idle and Active (Sensing, Processing and Connecting) Mode
 - Fast Wakeup from Standby to Active
- **Package**
 - AiP FCCSP: 8.8 x 12.4 mm², with 159 pins
- **Temperature Range**
 - Operating Junction Temperature: -40°C to 125°C

Applications

- Motion detection/Presence detection
- People tracking/counting
- Video doorbell/IP Camera
- Air conditioner/Refrigerators
- Drone/robot obstacle avoidance
- In-cabin sensing for left behind (LBC) systems

Description

The RS6240 is a single chip 60GHz 2T4R FMCW AiP MIMO radar sensor with 2.4GHz wireless connectivity (BLE5.3/802.15.4/Thread/Matter) inside. The device is portioned into four subsystems, shown in Figure below:

- **mmWave Radar Subsystem:** This block includes all high-performance radar RF/analog and baseband. It also includes the chirp generator and tx-to-rx finite state machine. HWA is included in baseband, used to offloading specified radar processing such as 1D/2D FFT and CFAR.
- **2.4GHz Wireless Connectivity Subsystem:** This block contains whole 2.4G multi-protocol wireless RF, Mod/Demod, Baseband Protocol, and the data interacts with application CPU by internal data bus.
- **Application Subsystem:** It mainly includes two 32bit RISC-V CPU cores with DSP, FPU and I/D cache. CPU connects SRAM, peripheral interface, and other subsystems by AHB and APB bus. RISC-V CPU is used to run radar post-processing algorithm, wireless protocol and other system tasks.
- **Power and Clock Subsystem:** This chip is complex single chip SoC with multiple function systems, we use single power and clock subsystem to do centralized management. It includes several clock sources (including LSI, MSI, Xtal Oscillator and SPLL) and power blocks(LDOs, BG and Clamp).

RS6240 is specifically designed for low power, high accuracy radar sense system applications. It contains four main power modes for low power management: shutdown, standby, idle and active. Its configuration and data acquisition are enabled with UART/SPI/I²C/CAN-FD/LIN digital interface with other devices.

Block Diagram

